

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	
2	Proper File Structure	Files and folders are logically organized	yes	
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	
4	Function / Method Names	Functions are descriptively named	Yes	
5	Code Comments	Inline and block comments explain logic	Partial	
6	Modular Design	Code is split into reusable functions/modules	Yes	
7	No Redundant Code	Duplicate or unused code is removed	Partial	
8	Error Handling	Exceptions and errors are handled gracefully	Yes	

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	scaler.pkl	Python / scikit-learn (StandardScaler)	Reused in app.py /submit endpoint to normalize new user inputs.	High
2	loan_model.pkl	Python / scikit-learn or XGBoost / joblib	Reused in app.py for prediction inference on new applications.	High
3	Missing Value Imputer Logic	Python / Pandas	Can be reused across any future tabular dataset requiring median/mode imputation.	High
4	Categorical Encoder (Dummy Vars)	Python / Pandas	Reused across the training pipeline and the active Flask application.	Medium
5	Flask Form Templates (predict.html)	HTML / Jinja2	The UI structure can be repurposed for other predictive forms (e.g., Credit Card Approval).	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure PDF	5	The separation of EDA, preprocessing, training, and the web app is a best practice.
Readability PDF	4	File names (<code>analysis.py</code> , <code>train_model.py</code>) immediately denote their function.
Reusability PDF	5	Persisting the model via <code>.pkl</code> files perfectly decouples the ML logic from the Flask app, maximizing reusability.
Documentation / Comments PDF	4	The project documentation provides a strong overview, but code-level inline comments should continually be maintained.
Overall Score PDF	4.5	Excellent modular design and deployment strategy.